

ENGLISH VERSION RESEARCH PROTOCOL

Thai Title

การจำแนกประเภทความผิดพลาดของใบสั่งยาแบบหลายป้ายกำกับด้วย Knowledge Base Generative AI

English Title

Multi-label Classification of Prescription Errors Using Knowledge-Based Generative AI

Principal Investigator: Mr. Siwawoot Mongkol

Position/Institution: Pharmacist, Pharmacy Department, Bangkok Hospital Khon Kaen

Academic Program: Master of Pharmacy Program in Health Informatics, Faculty of Pharmacy, Silpakorn University

Study Site: Bangkok Hospital Khon Kaen

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1. BACKGROUND AND RATIONALE

Prescription errors are an important patient-safety concern and may involve inappropriate drug selection, dose, frequency, route, duration, duplication, drug interactions, contraindications, allergy-related issues, or incomplete prescribing information. Pharmacist review can reduce these risks; however, reviewing complex prescriptions requires integration of multiple clinical variables and may be time-consuming, particularly when prescriptions contain several medications.

Generative artificial intelligence can process unstructured clinical text and identify relationships among medication-related data. Nevertheless, a stand-alone language model may generate inconsistent responses, unsupported explanations, or information that is not current. Linking the model to a curated and validated knowledge base may improve accuracy, consistency, traceability, and interpretability.

This study will develop and evaluate a Knowledge-Based Generative AI system for multi-label detection and classification of prescription errors. Its performance will be compared with a baseline generative AI system using the same underlying model without knowledge-base retrieval. Pharmacist assessment will serve as the reference standard. This is a retrospective study, and AI outputs will not be used for actual patient-care decisions.

2. STUDY OBJECTIVES

2.1 Primary objective

To develop and evaluate the performance of a Knowledge-Based Generative AI system for detecting and classifying prescription errors using a multi-label framework, compared with a pharmacist-derived reference standard.

2.2 Secondary objectives

- 1) To compare the performance of Knowledge-Based Generative AI with generative AI without knowledge-base retrieval.
- 2) To assess agreement between AI classifications and pharmacist assessments.
- 3) To analyze the characteristics and causes of discordant AI results to guide system improvement.

3. RESEARCH QUESTION AND HYPOTHESIS

The main research question is how accurately the Knowledge-Based Generative AI system can classify prescription errors using a multi-label approach compared with pharmacist assessment, and whether it performs better than generative AI without a knowledge base.

The study hypothesis is that Knowledge-Based Generative AI will achieve Precision, Recall, and F1-score values that are not inferior to, and are expected to exceed, those of the baseline generative AI system, particularly for error categories that depend on explicit clinical criteria or specialized reference information.

4. STUDY DESIGN AND SETTING

This is a retrospective observational study designed to evaluate diagnostic classification performance and validate an AI model. The study will use prescription and relevant clinical data obtained from the hospital information system of Bangkok Hospital Khon Kaen.

The retrospective data period will be from 1 May 2025 to 30 April 2026. The overall project period will extend from February to November 2026.

5. STUDY POPULATION, SAMPLE SIZE, AND SAMPLING

5.1 Target population

Prescriptions generated for patients receiving medication orders at Bangkok Hospital Khon Kaen during the specified study period.

5.2 Sample size

A minimum of 400 prescriptions will be included. The sample size is based on the Taro Yamane formula, assuming an estimated population of approximately 200,000 prescriptions and a 5% margin of error. Prescriptions excluded because of incomplete or non-processable data will be replaced using the same sampling method until at least 400 evaluable prescriptions are obtained.

5.3 Sampling method

Eligible prescriptions will be arranged chronologically by order date and time. After a random starting point is selected, systematic random sampling will be applied using a sampling interval calculated from the eligible population and required sample size.

6. ELIGIBILITY CRITERIA

6.1 Inclusion criteria

- 1) A prescription generated at Bangkok Hospital Khon Kaen between 1 May 2025 and 30 April 2026.
- 2) A prescription containing at least five medication items.
- 3) Availability of medication-order and clinical information required for assessment using the approved Case Record Form.
- 4) Removal or masking of direct identifiers before the data are transferred to the investigator.

6.2 Exclusion criteria

- 1) Duplicate records.
- 2) Records with insufficient essential information to evaluate the medication order.
- 3) Data corrupted or incorrectly formatted because of information-system or network failure.
- 4) Records that cannot be processed by the AI systems under the predefined data format.

7. STUDY VARIABLES AND DATA ELEMENTS

The study will use only information necessary for prescription assessment, including a research code, age, weight, height, blood pressure, body mass index, body surface area, drug and food allergy information, diagnoses or ICD codes, medication names, doses, directions, frequency, route, quantity or duration, and medication-order status.

The investigator will not receive or record patient names, national identification numbers, hospital numbers, visit numbers, order numbers, addresses, telephone numbers, or other direct identifiers. The investigator will not receive a re-identification key.

The outcome for each prescription will be multi-label. A prescription may contain no error, one error category, or multiple concurrent error categories. The final list of prescription-error labels and decision criteria will be defined in an expert-reviewed classification manual before the main analysis begins.

8. KNOWLEDGE BASE AND AI SYSTEMS

8.1 Knowledge base

The knowledge base will be developed from medication-use guidelines, hospital prescribing-review policies or criteria, approved product information, reliable drug-information references, and the study's prescription-error classification criteria. Content will be reviewed for accuracy, relevance, and currency before use. Each knowledge-base version and effective date will be documented.

8.2 Systems to be compared

- 1) Knowledge-Based Generative AI: a language model that retrieves relevant content from the approved knowledge base before generating classifications and explanations.
- 2) Baseline Generative AI: the same underlying language model using the same input data but without knowledge-base retrieval.

The two systems will use predefined and comparable prompts, input structures, output formats, and processing conditions. The model version, API version, parameters, prompt version, knowledge-base version, and processing date will be recorded and locked before the main analysis.

Only de-identified data will be submitted to the AI systems. No re-identification key will be transferred. Where supported by the service provider, settings that prevent submitted data from being used for additional model training will be selected. All use of the AI service must comply with hospital information-technology and data-protection policies.

9. PHARMACIST REFERENCE STANDARD

At least three licensed or clinical pharmacists from Bangkok Hospital Khon Kaen will independently review the selected prescriptions using the same approved classification manual. Reviewers will be blinded to AI results during their assessments and will record applicable prescription-error labels and supporting reasons.

Disagreements will initially be resolved by majority decision. Cases that remain unresolved will undergo adjudication through discussion and consensus. The final adjudicated result will serve as the reference standard for evaluation of both AI systems.

10. STUDY PROCEDURES

- 1) Obtain approval from the institutional ethics committee and authorization to use hospital data.
- 2) Authorized hospital personnel will extract data within the approved scope and remove direct identifiers before transfer to the investigator.
- 3) The investigator will check completeness, remove duplicate records, and select prescriptions according to the sampling plan.
- 4) The classification manual, knowledge base, prompt, and output format will be developed and pilot-tested using data not included in the main analytical dataset.
- 5) Pharmacists will independently assess prescriptions and establish the reference standard.
- 6) Prescriptions will be processed by both the Knowledge-Based Generative AI and baseline generative AI systems under predefined conditions.
- 7) Results will be entered in the Case Record Form, checked for quality, and compared with the reference standard.
- 8) Discordant cases will be analyzed, and findings will be summarized for the final report.

AI outputs will be used solely for research. They will not be entered into the medical record, communicated to the treating team, or used to modify patient treatment.

11. OUTCOME MEASURES

11.1 Primary outcomes

Precision, Recall, and F1-score of the Knowledge-Based Generative AI system compared with the pharmacist reference standard, reported using both micro-average and macro-average methods.

11.2 Secondary outcomes

- 1) Precision, Recall, and F1-score for each prescription-error category.
- 2) Agreement among pharmacist reviewers and between each AI system and the reference standard.
- 3) Exact-match ratio, defined as the proportion of prescriptions for which the complete predicted label set matches the reference label set.
- 4) Area Under the Curve, only when the AI system provides interpretable confidence scores and the threshold is prespecified.
- 5) Rates of processing failure, incomplete output, unsupported reasoning, and hallucination or inconsistency with input data or the approved knowledge base.

12. STATISTICAL ANALYSIS

Baseline characteristics will be summarized using number and percentage, mean and standard deviation, or median and interquartile range, as appropriate.

For each label, True Positive, False Positive, True Negative, and False Negative values will be determined. Precision, Recall, and F1-score will be calculated for each label and summarized using micro-average and macro-average methods. Ninety-five percent confidence intervals will be estimated using bootstrap methods when appropriate.

Agreement among pharmacists will be evaluated using Cohen's kappa or Fleiss' kappa, depending on the number of reviewers. Agreement between each AI system and the reference standard will be assessed using appropriate agreement statistics.

Because both AI systems will process the same prescriptions, paired analytical methods will be used. McNemar's test may be used for paired binary outcomes, and bootstrap methods will be used to estimate confidence intervals for differences in F1-score. Statistical significance will be set at 0.05.

When confidence scores are available, the threshold for classifying an output as positive or negative will be determined during pilot testing and locked before analysis of the main dataset. Records that cannot be assigned a reference standard because of incomplete information will be excluded from performance calculations and reported with reasons. Missing data will not be imputed.

13. QUALITY CONTROL

- 1) A single set of definitions and an approved classification manual will be used by all reviewers.
- 2) Pharmacist reviewers will receive orientation and competency checking before the main assessment.
- 3) Pilot-test data will be kept separate from the main analytical dataset.
- 4) The model version, prompt, knowledge base, and decision thresholds will be locked before the main analysis.
- 5) Data will be checked for duplicates, completeness, outliers, format errors, and direct identifiers before analysis.
- 6) Processing logs, system versions, and outputs will be retained to support traceability and reproducibility.

14. ETHICAL CONSIDERATIONS

14.1 Retrospective patient data

The study uses retrospective prescription and clinical data. There will be no contact with patients and no clinical intervention. Direct identifiers will be removed or masked before data are transferred to the investigator. A waiver of informed consent will therefore be requested because the study presents no more than minimal risk, obtaining individual consent would be impracticable, and the waiver will not adversely affect patient rights or welfare.

14.2 Pharmacist participants

Pharmacists who serve as study reviewers will be research participants. They will receive a Participant Information Sheet and provide written informed consent before participation. Participation will be voluntary, and refusal or withdrawal will not affect employment, performance appraisal, or professional relationships. The estimated total review workload is approximately 10–20 hours, divided into sessions of no more than about two hours to minimize fatigue. No compensation will be provided, and participants will incur no cost.

14.3 Risks and risk mitigation

The principal patient-data risk is breach of confidentiality. This risk will be minimized through de-identification, access restriction, password protection, encryption, and exclusion of any re-identification key from AI processing.

Risks to pharmacist participants include time burden, fatigue, and possible concern about individual performance. These risks will be minimized through flexible scheduling, rest periods, and assurance that individual review results will not be used for employment evaluation.

14.4 Benefits

Patients and pharmacist participants are not expected to receive direct benefit. The study may provide evidence for future development of tools that support prescription review and medication safety. The evaluated system will not be considered ready for clinical use unless it undergoes additional validation and approval under hospital governance processes.

15. DATA MANAGEMENT AND CONFIDENTIALITY

Research data will be stored on a password-protected and encrypted computer or storage environment approved by the hospital. Access will be restricted to the investigator and authorized personnel. Paper records, if any, will be kept in a locked cabinet. The investigator will not create or retain a linkage table to hospital numbers or other direct identifiers.

Data will be retained for five years after study completion. After the retention period, paper documents will be securely destroyed and electronic files will be permanently deleted in accordance with hospital policy. Results will be reported only in aggregate form. Raw data will not be disclosed to external parties without authorization.

16. STUDY TIMELINE

February–May 2026:	Literature review and refinement of the conceptual framework.
May–June 2026:	Development of prescription-error criteria, knowledge base, and research tools.
May–August 2026:	Preparation of documents and ethics submission.

September 2026:	Expert validity assessment and pilot testing.
September–October 2026:	Data extraction, cleaning, and sampling.
October 2026:	Pharmacist assessment, AI processing, and statistical analysis.
October–November 2026:	Interpretation, discussion, and thesis preparation.

17. DISSEMINATION OF RESULTS

Study findings may be included in a thesis, scientific conference presentation, or journal article, subject to hospital approval before dissemination. Only aggregate results will be reported, without disclosure of identifiable patient or pharmacist information or confidential internal hospital information.

18. CONFLICTS OF INTEREST AND FUNDING

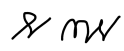
The investigator has no financial conflict of interest with the AI service providers or databases used in the study. The study has no external funding, and all expenses will be borne by the investigator.

19. STUDY DOCUMENTS

- 1) Case Record Form, Version 1.0.
- 2) Prescription Error Classification Manual and decision criteria.
- 3) Participant Information Sheet and Informed Consent Form for pharmacist reviewers.
- 4) Request for waiver of informed consent for retrospective patient data.
- 5) Authorization request for use of medical-record and hospital-information-system data.

20. KEY REFERENCES

- 1) World Health Organization. Medication Without Harm: Global Patient Safety Challenge on Medication Safety. Geneva: World Health Organization; 2017.
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- 3) Topol EJ. High-performance medicine: the convergence of human and artificial intelligence. *Nat Med.* 2019;25:44–56.
- 4) Singhal K, Azizi S, Tu T, et al. Large language models encode clinical knowledge. *Nature.* 2023;620:172–180.
- 5) Tsoumakas G, Katakis I. Multi-label classification: an overview. *Int J Data Warehous Min.* 2007;3(3):1–13.
- 6) Bishop CM. *Pattern Recognition and Machine Learning.* New York: Springer; 2006.

Principal Investigator's Signature: 

(Mr. Siwawoot Mongkol)

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